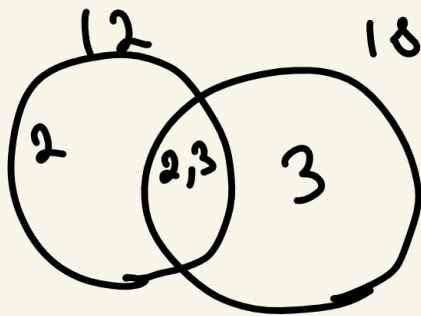
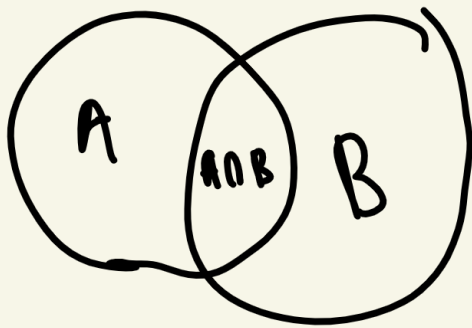



office hours
Erie 3125

January 5, 2026



$$\frac{ab}{\gcd(a,b)} = \text{lcm}(a,b)$$



$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proposition 1:

For all $x \in \mathbb{R}$, $x \cdot 0 = 0$

and $0 \cdot x = 0$

proof: Let $x \in \mathbb{R}$

$$x + 0 = x$$

$$1 + 0 = 1 \quad \text{by identity}$$

$$0 + 0 = 0$$

$$x(1 + 0) = x \cdot 1$$

$$x(1 + 0) = x \quad \text{by identity}$$

$$x \cdot 1 + x \cdot 0 = x \quad \text{by distributivity}$$

$$x + x \cdot 0 = x \quad \text{by identity}$$

$$-x + (x + x \cdot 0) = -x + x$$

$$-x + x + x \cdot 0 = -x + x \quad \text{Associative}$$

$$0 + x \cdot 0 = 0 \quad \text{invertibility}$$

$$x \cdot 0 = 0 \quad \text{identity}$$

$$\text{Also, } 0 \cdot x = 0 \quad \text{for all}$$

$$x \in \mathbb{R}, x \cdot 0 = 0 \text{ and } 0 \cdot x = 0$$


January 7, 2026

Proposition 2: For all $x, y \in \mathbb{R}$

$$(-x)y = -(xy)$$

proof: Let $x, y \in \mathbb{R}$

Universal Generalization

To prove "for all $x \in \mathbb{R}$, $P(x)$ "

Let $x \in \mathbb{R}$

* Demonstrate $P(x)$

Therefore, for all $x \in \mathbb{R}$, $P(x)$

$$-\square + \square = 0$$

$$-(xy) + xy = 0 \quad \text{by invertibility}$$

$$\underbrace{-(xy)}_{\text{keep}} + xy + (-x) \cdot y = 0 + \underbrace{(-x) \cdot y}_{\text{keep}}$$

$$-(xy) + xy + (-xy) = 0 + (-x) \cdot y$$

$$-(xy) + xy + (-1)(xy) = 0 + (-x) \cdot y$$

$$1x = x \quad - (1x) = -x \quad (-1)x = -x$$

$$-(xy) + (x + (-x))y = (-x)y \quad \begin{array}{l} \text{Distrib.} \\ \text{and Ident.} \end{array}$$

$$-(xy) + 0 \cdot y = (-x)y \quad \text{Invertibility}$$

$$-(xy) + 0 = (-x)y \quad \text{Prop 1}$$

$$-(xy) = (-x)y \quad \text{by identity}$$

Therefore, for all $x, y \in \mathbb{R}$ $(-x)y = -(xy)$

Proposition 3:

For all $x, y \in \mathbb{R}$, $(-x)(-y) = xy$

Proof: Let $x, y \in \mathbb{R}$

$$-x + x = 0 \quad \text{Invertibility}$$

$$(-x + x)(-y) = 0 \cdot (-y)$$

$$(-x)(-y) + x(-y) = 0 \cdot (-y) \quad \text{dist.}$$

$$(-x)(-y) + x(-y) = 0 \quad \text{Prop 1}$$

$$(-x)(-y) + x(-y) + xy = 0 + xy$$

$$(-x)(-y) + x(-y) + xy = xy \quad (\text{Ident.})$$

$$(-x)(-y) + x(-y+y) = xy \quad \text{dist.}$$

$$(-x)(-y) + x \cdot 0 = xy \quad \text{invertib.}$$

$$(-x)(-y) + 0 = xy \quad \text{Prop 1}$$

$$(-x)(-y) = xy \quad \text{Identity}$$

$$\text{For all } x, y \in \mathbb{R} \quad (-x)(-y) = xy$$

January 9, 2026

My dog is yellow
subject

X is yellow $\nearrow$ open sentence

Examples: $y = 2x + 1$
 $x < 3$

Each variable has an allowable
set of values called the

"universe of discourse" for variable

Example: X is wearing Y

Quantified Statements:

Two of my cats are orange

predicate: X is orange

Universe of discourse for x :

The set of all my cats

Universal Quantified Statements

"All my cats are orange"

predicate: X is orange

universe: C = set of my cats

Notation: $\forall x \in C, x \text{ is orange}$

Read: "for all values of x in C ,
 x is orange"

Existential Quantified Statements

"Some of my cats are orange"

Notation: $\exists x \in C, X \text{ is orange}$

Read: "for at least one value of x in C , x is orange"

"there is a value of x in C where x is orange"

"there exists an x in C for which x is orange"

$\mathbb{N}$ natural nums $\{1, 2, 3, 4, \dots\}$

$\mathbb{Z}$ integers $\{\dots -4, -3, \dots 0, 1, 2, 3, \dots\}$

$\mathbb{Q}$ rational nums All fractions & integ.

$\mathbb{R}$ real nums

$\mathbb{C}$ complex nums of real / imagin. numbers
combination

1. $\forall x \in \mathbb{N}$, $0 \leq x$ True

2. $\forall x \in \mathbb{R}$, $0 < x^2$ False

3. $\forall x \in \mathbb{R}$, $\exists y \in \mathbb{R}$, $x < y$ True

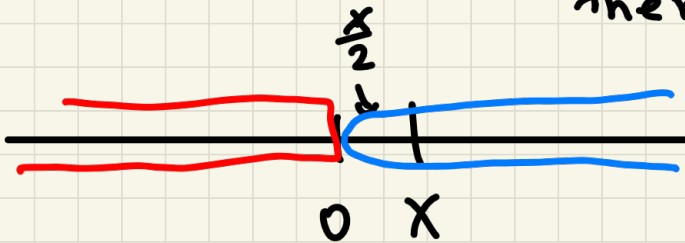
4. $\exists y \in \mathbb{R}$, $\forall x \in \mathbb{R}$, $x < y$ False

5. $\forall x \in \mathbb{Z}$, $\exists y \in \mathbb{Z}$, $y = 2x$ True

6. $\forall x \in \mathbb{Z}$, $\exists y \in \mathbb{Z}$, $x = 2y$ False
 $5 = 2y?$

7. $\forall x \in \mathbb{Z}$, If $\exists y \in \mathbb{Z}$, $x = 2y$ If its even then
 then $\exists q \in \mathbb{Z}$, $x^2 = 2q$ True

8. $\forall x \in \mathbb{R}$, If $\forall a \in (0, \infty)$, $x \leq a$
 then $x \leq 0$



If its $x \geq 0$, but never less or equal to zero

True

9. $\forall x \in \mathbb{R}$ If $\forall a \in \mathbb{R}$, $a \cdot x \leq 0$, {0}
 then $\forall b \in \mathbb{R}$, $0 \leq b \cdot x$ True

January 12, 2026

Proposition: $0 < 1$

Proof by contradiction:

To prove a proposition P

Assume $\neg P$ (negation of P)

derive a contrad. Q and $\neg Q$

Assume: $0 \neq 1$

Since $0 \neq 1$, we have $1 < 0$
by trichotomy

$-1 + 1 < -1 + 0$ by monotonicity

$$0 < -1$$

then:

$0(-1) < (-1)(-1)$ by monotonicity

$0 < 1$ by Prop 1 & Prop 3

Now, $0 < 1$ and $0 \leq 1$. This is contradiction

Therefore, $0 < 1$

Proposition: $1+1 \neq 1$

Proof:

Assume $1+1 = 1$

$$1+1(-1) = 1+(-1)$$

$$1 = 0$$

But, $1 \neq 0$ This is contradiction

Therefore, $1+1 \neq 1$

Proposition: $1+1 \neq 0$

Proof:

Assume $1+1 = 0$

We know $0 < 1$

then $1+0 < \underline{1+1}$ by monotonicity

So, $1 < 0$

this is a contradiction, since $0 < 1$

Therefore, $1+1 \neq 0$

Definition:

$$2 = 1+1 \quad 3 = 2+1 \quad 4 = 3+1 \quad 5 = 4+1$$

Example:

$$2 < 4$$

Proof:

$$0 < 1$$

$$\text{then } 1+0 < 1+1$$

$$\text{so } 1 < 2$$

by transitivity, $0 < 2$

$$\text{then } 1+0 < 2+1$$

$$\text{so } 1 < 3$$

$$\text{then } 1+1 < 3+1, \text{ so } 2 < 4$$

January 14, 2026

Proposition:

$\forall x, y \in \mathbb{R}$, if $x < y$, then $-y < -x$

Direct Proof

To prove If P , then Q

Assume P

* Demonstrate Q

Therefore, If P , then Q

Proof

Let $x, y \in \mathbb{R}$

Assume $x < y$

by monotonicity

$$-x + x + (-y) < -x + y + (-y)$$

$$0 + (-y) < -x + 0$$

$$-y < -x \quad \therefore$$

Therefore, If $x < y$, then $-y < -x$

Monotonicity (Negative Multiplication)

$\forall x, y, z \in \mathbb{R}$, If $x < y$ and $z < 0$,
then $yz < xz$

Proof Let $x, y, z \in \mathbb{R}$ target $yz < xz$

Assume $x < y$ and $z < 0$

Since $z < 0$, we have $0 < -z$
 $-z + z < -z + 0$

Then $x(-z) < y(-z)$

So, $-xz < -yz$

$$xz + (-xz) + yz < xz + (-yz) + yz$$

$$0 + yz < xz + 0$$

$$yz < xz$$

Therefore, if $x < y$ and $z < 0$,

QED then $yz < xz$

$\forall x, y \in \mathbb{R}$, if $x < y$, then $x+2 < y+3$

Proof Let $x, y \in \mathbb{R}$

Assume $x < y$

$$x+2 < y+3$$

$x+2 < y+2$ by monotonicity

Since $2 < 3$, $y+2 < y+3$ by monotonicity

$$x+2 < y+2 \text{ and } y+2 < y+3$$

By transitivity, $x+2 < y+3$

Using transitivity to Prove $A < B$

$$\textcircled{1} \quad A < C$$

$$\textcircled{2} \quad C < B$$

$$A < B$$

January 16, 2026

Rules of Negation

1. $\neg(\forall x \in U, P(x))$ is $\exists x \in U, \neg P(x)$
2. $\neg(\exists x \in U, P(x))$ is $\forall x \in U, \neg P(x)$
3. $\neg(P \text{ and } Q)$ is $\neg P \text{ or } \neg Q$
4. $\neg(P \text{ or } Q)$ is $\neg P \text{ and } \neg Q$
5. $\neg(\text{If } P, \text{ then } Q)$ is $P \text{ and } \neg Q$

$\forall x \in \mathbb{R}, \text{ If } \boxed{\overset{P}{\forall a \in (0, \infty), a \leq x}}, \text{ then } \boxed{\overset{Q}{x \leq 0}}$

Negation

$\exists x \in \mathbb{R}, \boxed{\overset{P}{\forall a \in (0, \infty), a \leq x}} \text{ and } \boxed{\overset{Q}{0 < x}}$

1. $\forall x \in \mathbb{Z}$, If $\exists a \in \mathbb{Z}, x = 2a + 1$, then
 $\exists b \in \mathbb{Z}, x = 3b$ 2

$\exists x \in \mathbb{Z}, \exists a \in \mathbb{Z}, x = 2a + 1$ and
 $\forall b \in \mathbb{Z}, x \neq 3b$ ✓

2. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x < y$

$\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, x \geq y$ ✓

3. $\forall x, y \in \mathbb{R}$, If $0 < x < 1$ and
 $x + y = 1$, then $0 < y < 1$

$\exists x, y \in \mathbb{R}, 0 < x < 1$ and $x + y = 1$ ✓
and $y \leq 0$ or $1 \leq y$

1. $A < C$
2. $C < B > A < B$

1. $\forall x, y \in \mathbb{R}$, If $0 < y < x$, then $y < 2x$

Structure Let $x, y \in \mathbb{R}$

Assume $0 < y < x$

$$0 < y \text{ and } y < x$$

$$\text{then } 0 < x \quad x+0 < x+y$$

$$\text{so } x < 2x$$

$$y < x < 2x$$

So

$$y < 2x$$

2. $\forall x, y \in \mathbb{R}$, If $x < 2 < y$, then $x+2 < y^2$

$$x < 2 \text{ and } 2 < y$$

$$2 \cdot 2 < 2 \cdot y \text{ so } 4 < 2y$$

Since $0 < 2 < y$
we have $0 < y$

$$2 < y \text{ so } 2(y) < y(y)$$

$$2y < y^2$$

$$\text{then } 4 < y^2$$

$$x < 2 \text{ so } x+2 < 4$$

$$\text{Now, } x+2 < 4$$

January 19, 2026

Proving Statements with Existential Qualifiers

Universal Generalization

To prove $\forall x \in \mathbb{R}, P(x)$

Let $x \in \mathbb{R}$

* Show $P(x)$

Therefore, $\forall x \in \mathbb{R}, P(x)$

Existential Generalization

To prove $\exists x \in \mathbb{R}, P(x)$

Let $x = \square$  some specific value

* Demonstrate $P(x)$

for that specific value

Therefore, $\exists x \in \mathbb{R}, P(x)$

Example

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, \exists y \in \mathbb{R}, x < y$$

proof: let $x \in \mathbb{R}$
let $y = x+1$

Since $0 < 1$, $x+0 < x+1$
then $x < y$

Therefore, $\exists y \in \mathbb{R}, x < y$

Example

$\forall x, y \in \mathbb{R}$, If $x < y$, then $\exists z \in \mathbb{R}$
 $x < z < y$

Proof:

let $x, y \in \mathbb{R}$

Assume $x < y$

$$\text{let } z = \frac{x+y}{2}$$

Since $x < y$

$$\text{then } x+x < y+x$$

$$2x < x+y$$

$$\frac{1}{2}(2x) < \frac{1}{2}(x+y)$$

$$\text{So, } x < z$$

Also, since $x < y$, $x+y < y+y$

$$\text{then } x+y < 2y$$

$$\text{so } 2^{-1}(x+y) < 2^{-1}2y$$

$$\text{Now } \frac{x+y}{2} < y, \text{ so } z < y$$

Therefore, $\exists z \in \mathbb{R}, x < z < y$

Example

$\forall x \in \mathbb{R}$, If $2 < x$, then
 $\exists a \in \mathbb{R}, 1 < a$ and $1+a < x$

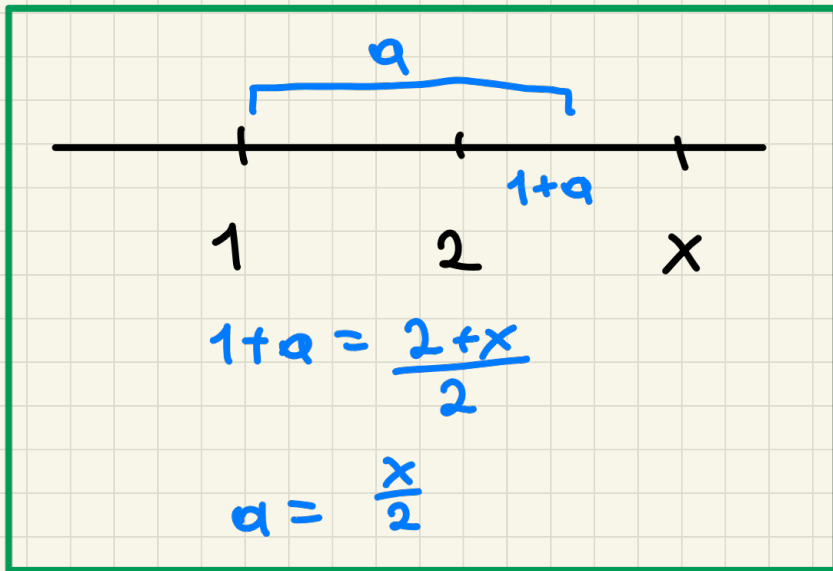
Proof

Let $x \in \mathbb{R}$

Assume $2 < x$

Let $a = x/2$

$1 < a$



Since $2 < x$, $\frac{1}{2}(2) < \frac{x}{2}$

So, $1 < a$

Then, $1+a < a+a$

So, $1+a < 2a$

$1+a < 2\left(\frac{x}{2}\right)$ so $1+a < x$

Therefore $1 < a$ and $1+a < x$

So, $\exists a \in \mathbb{R}$, $1 < a$ and $1+a < x$

Example

$\forall x, y \in \mathbb{R}$, If $0 < x < 1$ and $0 < y < 1$,

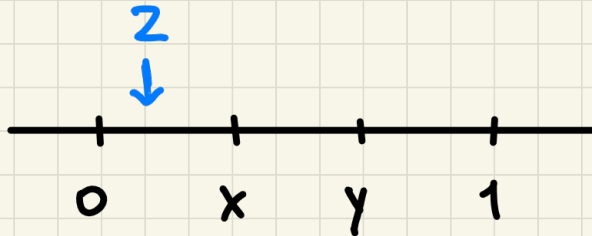
then $\exists z \in \mathbb{R}$, $0 < z < x$ and $0 < z < y$

Proof

let $x, y \in \mathbb{R}$

Assume $0 < x < 1$ and $0 < y < 1$

let $z = x \cdot y$



TBC

January 21, 2026

$\wedge$ "and"

$\wedge$	T	F
T	T	F
F	F	F

$\vee$ "or"

$\vee$	T	F
T	T	F
F	T	F

$B = \{T, F\}$

Inclusive OR - includes the possibility of both statements being true

$\neg$ "not" $\neg F = T$ $\neg T = F$

$x \Rightarrow y$ statement or binary operation

In COMP 1000 $x \Rightarrow y$ is the same as $\neg x \vee y$ but not MATH 1020

In IR $x(y+z) = xy + xz$

in $\mathbb{B}$ boolean

$$x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$$

$$x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$$

$$x \wedge (x \vee y) = x$$

$$x \vee (x \wedge y) = x$$

$x \Rightarrow y$ on $\mathbb{B}$

behaves analogous to $x \leq y$ on $\mathbb{R}$

$$x \wedge T = x$$

$$x \vee F = x$$

$$x \vee T = T$$

$$x \wedge F = F$$

January 23, 2026

Identity:

$$x \vee F = x$$

$$x \wedge T = x$$

Annihilation:

$$x \wedge F = F$$

$$x \vee T = T$$

Complementation:

$$x \wedge \neg x = F \quad \neg \boxed{\neg x} \wedge \boxed{\neg x} = F$$

$$x \vee \neg x = T \quad \neg \boxed{\neg x} \vee \boxed{\neg x} = T$$

Proposition (Involution)

$$\forall x \in B, \neg \neg x = x$$

Proof:

$$\text{let } x \in B$$

$$x \vee F = x \quad (\text{Identity})$$

$$x \vee (\neg \neg x \wedge \neg x) = x \quad (\text{complementation})$$

$$(x \vee \neg \neg x) \wedge (x \vee \neg x) = x \quad (\text{Distributivity})$$

$$(x \vee \neg \neg x) \wedge \underline{T} = x \quad (\text{Complementation})$$

$$(x \vee \neg \neg x) \wedge \underline{(\neg \neg x \vee \neg x)} = x$$

$$\neg \neg x \vee (x \wedge \neg x) = x$$

$$\neg\neg x \vee F = x$$

So, $\neg\neg x = x$ By identity

$$\textcircled{1} \forall x, y \in B, (x \wedge \neg y) \vee (y \wedge \neg x) = (x \vee y) \wedge (\neg x \vee \neg y)$$

$$\textcircled{2} \forall x, y \in B, \text{ If } x \wedge \neg y = x, \text{ then}$$

$$\neg x \wedge y = y$$

$\textcircled{1}$

let $x, y \in B$

$$(x \wedge \neg y) \vee (y \wedge \neg x) = (x \vee y) \wedge (\neg x \vee \neg y)$$

$$(x \wedge y) \vee \underbrace{(x \wedge \neg x)}_F \vee \underbrace{(\neg y \wedge y)}_F \vee (\neg y \wedge \neg x)$$

$$(x \wedge y) \vee (\neg y \wedge \neg x) = (x \vee y) \wedge (\neg x \vee \neg y)$$

$$\neg(x \wedge y) \wedge \neg(\neg y \wedge \neg x) \quad \text{DM}$$

$$(\neg x \vee \neg y) \wedge (y \vee x)$$

$$(y \vee x) \wedge (\neg x \vee \neg y) = (x \vee y) \wedge (\neg x \vee \neg y)$$

②

let $x, y \in B$

assume $x \wedge \neg y = x$ $\neg x \wedge y = y$

$$\neg(x \wedge \neg y) = \neg x$$

$$\neg x \vee \neg \neg y = \neg x$$

$$\neg x \vee y = \neg x \quad | \wedge y$$

$$(\neg x \vee y) \wedge y = \neg x \wedge y \quad \text{Absorption}$$

$$\underline{y = \neg x \wedge y}$$

Axioms of a binary operation

Existence: $\forall x, y \in \mathbb{R}, x + y \in \mathbb{R}$

Uniqueness: $\forall a, b, x \in \mathbb{R}, \text{ if } a = b, \text{ then}$
 $a + x = b + x$

Existence: $\forall x, y \in B, x \wedge y \in B$

Uniqueness: $\forall a, b, x \in B, \text{ if } a = b, \text{ then}$
 $a \wedge x = b \wedge x$

③

$\forall x, y, a \in B, \text{ if } x \Rightarrow y \text{ and}$
 $a \Rightarrow b, \text{ then } x \wedge a \Rightarrow y \wedge b$

let $x, y, a \in B$

Assume $x \Rightarrow y$ and $a \Rightarrow b$

$$x \wedge a \Rightarrow y \wedge a$$

$$x \wedge a \Rightarrow y \wedge a$$

$$y \wedge a \Rightarrow y \wedge b$$

$$x \wedge y \Rightarrow y \wedge b$$

Since $a \Rightarrow$, $y \wedge a \Rightarrow y \wedge b$

By transitivity, $x \wedge a \Rightarrow y \wedge b$

January 26, 2026

Proposition: For all $x, y \in B$, the following statements are equal

(1) $x \Rightarrow y$

(2) $x \wedge y = x$

(3) $x \vee y = y$

(4) $\neg x \vee y = T$

Proof: let $x, y \in B$

(1) $\Rightarrow$ (2)

Assume $x \Rightarrow y$ $x \wedge y = x$

$x \wedge y \Rightarrow x$ by consistency

Since $x \Rightarrow y$, $x \wedge x \Rightarrow x \wedge y$

Then $x \Rightarrow x \wedge y$

$x \wedge y = x$ by antisymmetry

Therefore, if (1) then (2)

(2) $\Rightarrow$ (3)

Assume $x \wedge y = x$ $x \vee y = y$

Then $(x \wedge y) \vee y = x \vee y$

By Absorption, $y = x \vee y$

Therefore, if (2) then (3)

(3) $\Rightarrow$ (4)

Assume $x \vee y = y$ $\neg x \vee y = T$

$$\frac{T}{\neg x \vee x \vee y} = \neg x \vee y \quad \vee \neg x$$

$$T \vee y = \neg x \vee y$$

Then, $T = \neg x \vee y$

If (3) then (4)

(4) $\Rightarrow$ (1)

Assume $\neg x \vee y = T$ $x \Rightarrow y$

$$x \wedge (\neg x \vee y) = x \wedge T \quad \wedge x$$

$$(x \wedge \neg x) \vee (x \wedge y) = x$$

$$F \vee (x \wedge y) = x$$

$$x \wedge y = x$$

Also, $x \wedge y \Rightarrow y$ by consistency

Then $x \Rightarrow y$
Therefore, If (4) then (1)

January 30, 2026

Proving Conditional Statements

To prove "If A , then B "

Direct Proof $A \Rightarrow B$
Assume $A \wedge B = A$
Assume $A \vee B = B$

↓
Proposition (Contraposition)
* demon

$\forall x, y \in B, x \Rightarrow y$ If and only If
 $\neg y \Rightarrow \neg x$

Proof: let $x, y \in B$
Assume $x \Rightarrow y$ $\neg y \Rightarrow \neg x$

Then $x \wedge y = x$ meet formula

So, $\neg y \vee \neg x = \neg x$
 $A \quad B \quad B$ join formula

Therefore, $\neg y \Rightarrow \neg B$

Conversely, assume $\neg x \Rightarrow \neg y$

Then, $\neg y \wedge \neg x = \neg y$ meet

So, $\neg\neg y \wedge \neg\neg x = \neg\neg y$

$y \vee x = y$ Involution

Then, $x \vee y = y$

$x \Rightarrow y$ join

January 30, 2026

Proving Conditional Statements

Direct Proof

Assume A

↓

* Demonstrate B

Therefore, If A
then B

Contraposition

Assume $\neg B$

↓

* Demonstrate $\neg A$

Therefore, If A
then B

① $\forall x, y \in \mathbb{R}$, If $y^2 \leq x+2$, then
 $2 \leq x$ or $y \leq 2$

② $\forall x, y, z \in \mathbb{R}$, If $xz + yz \leq x^2 + y^2$
then $x \leq 0$ or $y \leq x$ or $z \leq y$

③ $\forall x, y \in \mathbb{R}$, If $\forall a \in (-\infty, x-a)$,
then $y \leq x$

Proof:

1. Assume $x < 2$ and $2 < y$... $x+2 < y^2$

Since $x < 2$, $x+2 < 4$

Since $0 < 2 < y$, we

know $2 < y$

Now, $2 \cdot y < y \cdot y = 2y < y^2$

Since, $2 < y$, $x+2 < 2y$

2. Assume $0 < x < y < z$... $x^2 + y^2 < xz + yz$

Then $x < z$, so $x^2 < xz$

Then, $x^2 + yz < xz + yz$

Also, $y < z$ and $0 < y$, so $y^2 < yz$

Then, $x^2 + y^2 < x^2 + yz$

By transitivity, $x^2 + y^2 < xz + yz$

③ proof: Let $x, y \in \mathbb{R}$

Assume $x < y \dots \exists a \in (-\infty, 0), x - y < a$

$$\text{Let } a = \frac{x - y}{2}$$

$2a = x - y$

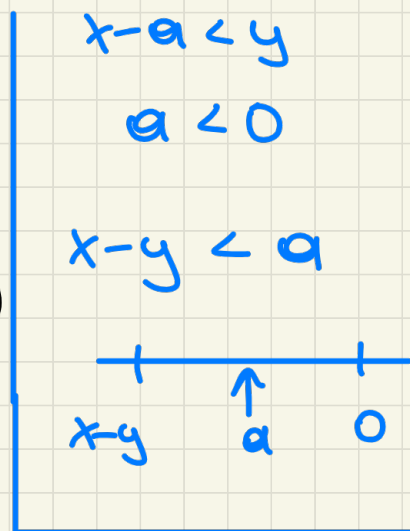
Since $x < y$ $x - y < 0$

Then $\frac{x - y}{2} < 0$, so $a \in (-\infty, 0)$

Since $a < 0$, $2a < a$

So, $x - y < a$

Then $x - a < y$



February 2, 2026

Instantiations (gathering information from your assumption)

Example

$$\forall x \in \mathbb{R}, ax \leq x \quad \text{then} \quad x = 0$$

Proof:

$$\text{Let } x \in \mathbb{R}$$

$$\text{Assume } \forall a \in \mathbb{R}, ax \leq x$$

$$\square \cdot x \leq x$$

$$\text{Since } 0 \in \mathbb{R}, 0 \cdot x \leq x$$

$$\text{Then } \underline{0 \leq x}$$

$$\text{Since } 2 \in \mathbb{R}, 2 \cdot x \leq x$$

$$\text{Then } 2x - x \leq 0, \underline{x \leq 0}$$

$$\text{Thus, } \underline{x = 0}$$

Example

$\forall x \in \mathbb{R}$, If $\forall a \in (1+x, \infty)$,
 $2x \leq 2+a$, then $x \leq 4$

Proof:

Let $x \in \mathbb{R}$

Assume $\forall a \in (1+x, \infty)$, $2x \leq 2+a$ ^{$x \leq 4$}

$$2x \leq 2 + \square$$

Since $1+x < 2+x$, we know
that $2+x \in (1+x, \infty)$

Universal
Instantiation

Then $2x \leq 2 + (2+x)$

$$2x \leq 4+x$$

$$2x - x \leq 4, \quad \underline{x \leq 4}$$

Then $x \leq 4$

Example

$\forall x \in \mathbb{R}$, If $\exists a \in \mathbb{R}$, $a \neq 1$
and $ax = x$, then $x = 0$

Proof

Let $x \in \mathbb{R}$

Assume $\exists a \in \mathbb{R}$, $a \neq 1$ and $ax = x$ $x = 0$

Existential Instantiation

Choose $\alpha \in \mathbb{R}$ with $\alpha \neq 1$ and $\alpha x = x$

$$\alpha x - x = 0$$

$$(\alpha - 1)x = 0$$

Since $\alpha \neq 1$, we know $\alpha - 1 \neq 0$

$$(\alpha - 1)^{-1} (\alpha - 1)x = (\alpha - 1)^{-1} 0$$

$$\underline{x = 0}$$

Example

$\forall x \in \mathbb{R}$, If $\exists b \in (0, \infty)$, $\forall a \in (0, \infty)$
 $bx < a$, then $\forall c \in (0, \infty)$, $x < c$

Proof

Let $x \in \mathbb{R}$

Assume $\exists b \in (0, \infty)$, $\forall a \in (0, \infty)$ $bx < a$

Let $c \in (0, \infty)$

Choose $\beta \in (0, \infty)$, where $\forall a \in (0, \infty)$, $\beta x < a$

Since $\beta \in (0, \infty)$ and $c \in (0, \infty)$,
we know $\beta c \in (0, \infty)$

Then $\beta x < \beta c$, then $x < c$

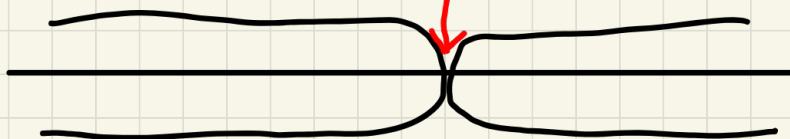
February 4, 2026

Theorem (The Archimedean Property)

$\forall a, b \in \mathbb{R}$, If $0 < a$, then $\exists n \in \mathbb{N}$, $b < na$



$$A = \{x \in \mathbb{Q} \mid 0 < x, x^2 < 2\} \quad B = \{x \in \mathbb{Q} \mid 0 < x, 2 < x^2\}$$



Dedekind Cut

$$c = \sqrt{2}$$

Proof:

$$\text{Let } a, b \in \mathbb{R}$$

$$\text{Assume } 0 < a$$

$$\forall n \in \mathbb{N}, a_n \leq b$$

reachable Let $A = \{a_n \mid n \in \mathbb{N}\}$

unreachable Let $B = \{y \in \mathbb{R}, \forall n \in \mathbb{N}, a_n \leq y\}$

Since $a \in A$, $b \in B$, A and B are not empty

Also, for any $y \in B$ and any $x \in A$

$x = an$ for some $n \in \mathbb{N}$, and so $an \leq y$

Then A and B form a Dedekind cut

Then there is a number

February 6, 2026

The Archimedean Property

$\forall a, b \in \mathbb{R}$, if $0 < a$, then $\exists n \in \mathbb{N}$, $b < an$

Ex.

$\forall \epsilon \in \mathbb{R}$, if $0 < \epsilon$, then $\exists n \in \mathbb{N}$, $\frac{1}{n} < \epsilon$

Proof:

Let $\epsilon \in \mathbb{R}$

Assume $0 < \epsilon$

By the Archimedean Property

$\exists n \in \mathbb{N}$, $1 < \epsilon n$

Choose $n \in \mathbb{N}$ with $1 < \epsilon n$

$$\text{Then } \frac{1}{n} < \varepsilon$$

$$\text{Therefore, } \exists n \in \mathbb{N}, \frac{1}{n} < \varepsilon$$

Ex. $\forall x \in \mathbb{R}$, if $x < 1$, then $\exists n \in \mathbb{N}, x < \frac{n}{n+1}$

Proof:

$$\text{Let } x \in \mathbb{R}$$

$$\text{Assume } x < 1 \Rightarrow 0 < 1-x$$

By the Archimedean Property

$$\exists n \in \mathbb{N}, x < (1-x)n$$

$$\text{Choose } n \in \mathbb{N} \text{ with } x < (1-x)n$$

scrap paper

$$x < \frac{n}{n+1}$$

$$(n+1)x < n$$

$$nx + x < n$$

$$x < n - nx$$

$$x < (1-x)n$$

$$x < n - xn$$

$$xn + x < n$$

$$x(n+1) < n$$

$$x < \frac{n}{n+1}$$

therefore

$$\exists n \in \mathbb{N}, x < \frac{n}{n+1}$$

① $\forall x \in \mathbb{R}, \text{ if } 5 < x, \text{ then } \boxed{\exists n \in \mathbb{N}, \frac{5n+7}{n+1} < x}$

② $\forall x, y \in \mathbb{R}, \text{ if } x < y, \text{ then } \exists n \in \mathbb{N}, (3+n)x < (n+3)y$

③ $\forall x \in \mathbb{R}, \text{ if } \forall n \in \mathbb{N}, \frac{3n+2}{n+1} \leq x, \text{ then } 3 \leq x$

1) Let $x \in \mathbb{R}$

Assume $5 < x \Rightarrow 0 < x-5$

By the Archimedean Property

$\exists n \in \mathbb{N}, 7-x < (x-5)n$

Choose $n \in \mathbb{N}$ with $\uparrow$

$7-x < xn-5n$

$5n+7 < xn+x$

$5n+7 < x(n+1)$

$\boxed{\frac{5n+7}{n+1} < x}$

$\frac{5n+7}{n+1} < x$

$5n+7 < x(n+1)$

$5n+7 < nx+x$

$5n-nx+7 < x$

$n(5-x)+7 < x$

$x > n(5-x)$

$x > \boxed{5}n+7$

2) Let $x, y \in \mathbb{R}$

Assume $x < y$

By the Archimedean Property

$$\exists n \in \mathbb{N}, 3x + 3y < (y-x)n$$

Choose $n \in \mathbb{N}$ with $3x + 3y < (y-x)n$

$$3x + 3y < ny - nx$$

$$3x + xn < ny - 3y$$

$$\exists n \in \mathbb{N}, (3+n)x < (n-3)y$$

3) Let $x \in \mathbb{R}$

Assume $x > 3$
 $x-3 > 0$

$$x-2 < n(3-x)$$

$$x-2 < 3n - nx$$

$$3n+2 > xn+x$$

$$3n-xn > x-2$$

$$n(3-x) > x-2$$

$$n > (x-2)/(3-x)$$

$$x + nx < 3n + 2$$

$$x(1+n) < 3n+2$$

$$x < \frac{3n+2}{1+n}$$

February 11, 2026

The well-ordering property

If a non-negative integer exists with a certain property then a smallest non-negative number exists with that property,

Why?

Suppose we know $\exists x \in \mathbb{Z}, 0 \leq x, P(x)$

Consider the set $A = \{a \in \mathbb{Z}, 0 \leq a, P(a)\}$

The statement $\exists x \in \mathbb{Z}, 0 \leq x$ and $P(x)$

means A is not empty

By the well-ordering property

A has a smallest element

Existential Instantiation

Recall: If we assume $\exists a \in \mathbb{R}, P(a)$

Let $\alpha \in \mathbb{R}$ with $P(\alpha)$

Existential Instantiation with
the well-ordering property

If we assume $\exists a \in \mathbb{Z}, 0 \leq a$ and $P(a)$

Let $\alpha \in \mathbb{Z}$ be smallest for
which $0 \leq a$ and $P(a)$

Proposition

$\forall x \in \mathbb{Z},$ if $0 < x$, then $1 \leq x$

Proof:

Assume $\exists x \in \mathbb{Z}$, $0 < x$ and $x < 1$

Let $\alpha \in \mathbb{Z}$ be smallest with $0 < \alpha < 1$

Then $0\alpha < \alpha^2 < \alpha$

Also, $\alpha < 1$

So, $0 < \alpha^2 < \alpha < 1$

Now, $0 < \alpha^2 < 1$, but $\alpha^2 < \alpha$

This contradicts α being smallest

Therefore, $\forall x \in \mathbb{Z}$, if $0 < x$, then $1 \leq x$

Proposition

$\forall x \in \mathbb{R}$, if $0 < x$, then $\forall n \in \mathbb{N}$, $0 < x^n$

Proof:

Let $x \in \mathbb{R}$

Assume $0 < x$ and $\exists n \in \mathbb{N}$, $x^n \leq 0$

Let $m \in \mathbb{N}$ be smallest with $x^m \leq 0$

Since $0 < x^1$ but $x^m \leq 0$, we know $m \neq 1$

Then $m-1 \in \mathbb{N}$. Then $0 < x^{m-1}$

Now, $0 \cdot x < x^{m-1} \cdot x$

$0 < x^m$ but $x^m \leq 0$

This is a contradiction ■

Well ordering Principal

Archimedean:

$$\begin{array}{lcl} \text{any } a, b \in \mathbb{R} & & 0 < a \\ m \in \mathbb{N} & & b < m \cdot a \end{array}$$

1) $\forall x \in \mathbb{R}$, if $\exists m \in \mathbb{N}$, $x^m < 0$, then $x < 0$

Proof: Contraposition

$$\text{If } x \geq 0, \text{ then for all } m \in \mathbb{N}, \\ x^m \geq 0$$

Suppose $x \geq 0$

Assume (for a contradiction)
there exists $k \in \mathbb{N}$ such $x^k < 0$

Consider: $\mathcal{S} = \{ a \in \mathbb{N} \mid x^a < 0 \}$

By assumption, since $k \in \mathcal{S}$
 $\mathcal{S} \neq \emptyset$

By well ordering principle

there exists a minimal element

$$m \in \mathcal{S}$$

$$\forall x \in \mathbb{R}, \text{ if } \exists m \in \mathbb{N}, x^m < x, \text{ then } x < 1$$

Proof: Let $x \in \mathbb{R}$

Contraposition:

$$\text{if } x \geq 1, \text{ then for } \forall m \in \mathbb{N} \\ x^m \geq x$$

Suppose $x \geq 1$

$$\text{Assume } k \in \mathbb{N}, x^k < x$$

Archimedean Property

$$\forall x, y \in \mathbb{R}, \text{ if } 0 < \varepsilon, \text{ then } \exists m \in \mathbb{N}$$

$$\frac{mx+y}{m} < x + \varepsilon$$

Let $x, y, \epsilon \in \mathbb{R}$

Suppose $0 < \epsilon$

If , then $\forall_{n \in \mathbb{N}}$
well ordering

If , then $\exists_{n \in \mathbb{N}}$
archimedean prop.

March 2, 2026

Induction

Let S be non-empty set

S is called an "inductive" set when it has property:

$$\forall_{x \in S}, x+1 \in S$$

Example $(0, \infty)$ is inductive

$$\forall x \in (0, \infty), x+1 \in (0, \infty)$$

Proof

$$\text{Let } x \in (0, \infty) \dots x+1 \in (0, \infty)$$

$$0 < x$$

$$\text{Then, } 1 < x+1$$

$$\text{So, } 0 < x+1$$

$$\text{Thus, } x+1 \in (0, \infty)$$

Example

$$A = \{ x \in \mathbb{R} \mid 1 < x+1 < x^2 \}$$

A is inductive

Proof

$$\text{Let } x \in A \dots x+1 \in A$$

$$\text{Then, } 1 < x+1 < x^2$$

$$1 < \square + 1 < \square^2$$

Since $1 < x+1 < x^2$

$$1 < x+1, \quad 2 < (x+1)+1$$

$$\text{So } 1 < (x+1)+1$$

Target: $x+1 \in A$

$$1 < (x+1)+1 < (x+1)^2$$

$$1 < x+1, \text{ so } 0 < x$$

$$\text{Then } 0 < 2x \text{ so } x+1 < 3x+1$$

$$\text{Now, } x+1 < x^2 \text{ so } 3x < x^2+2x$$

$$\text{By transitivity, } x+1 < x^2+2x$$

$$x+1+1 < x^2+2x+1$$

$$(x+1)+1 < (x+1)^2$$

$$\text{Now, } 1 < (x+1)+1 < (x+1)^2$$

$$x+1 \in A$$

Example

$$B = \{t \in \mathbb{R}, \exists m, n \in \mathbb{Z}, t^2 = mn + n\}$$

$$\square \in \mathbb{R}, \exists m, n \in \mathbb{Z}, \square^2 = m\square + n$$

B is inductive

Proof

$$\text{Let } x \in B \quad \dots \quad x+1 \in B$$

$$\text{This means } \exists m, n \in \mathbb{Z}, x^2 = mx + n$$

$$\text{Target: } \exists m, n \in \mathbb{Z}, (x+1)^2 = m(x+1) + n$$

$$x+1 \in B$$

$$\text{Let } a, b \in \mathbb{Z} \text{ with } x^2 = ax + b$$

$$\text{Let } m = a+2$$

$$n = b - a - 1$$

$$(x+1)^2 = \underline{x^2} + 2x + 1$$

$$= \underline{ax + b} + 2x + 1$$

$$= (a+2)x + (b+1)$$

$$= (a+2)x + a+2 - a-2 + b+1$$

$$= (a+2)(x+1) + (b-a-1)$$

$$= m(x+1) + n$$